

Systems of many forms with differing degrees

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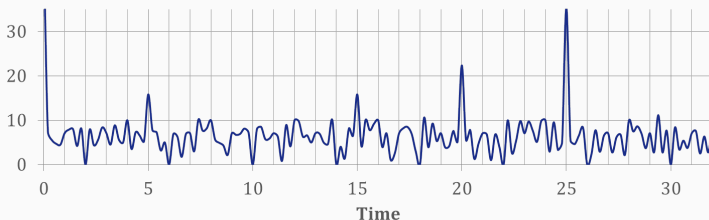
maths.fan/IML.pdf

Notation

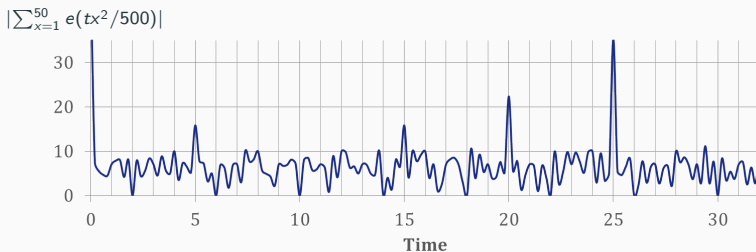
- $\vec{f}(\vec{x}) \in \mathbb{Z}[\vec{x}]^R$ is a system of R forms in s variables of degrees $d_i \geq 2$.
- Set $N_{\vec{f}}(B) := \#\{\vec{x} \in \mathbb{Z}^s : \vec{f}(\vec{x}) = \vec{0}, \|\vec{x}\| \leq B\}$, $\vec{\alpha} \cdot \vec{f} = \sum^R \alpha_i f_i$.

$$N_{\vec{f}}(B) = \int_0^1 \cdots \int_0^1 \sum_{\substack{\vec{x} \in \mathbb{Z}^s \\ \|\vec{x}\| \leq B}} e^{2\pi i \vec{t} \cdot \vec{f}(\vec{x})} d\vec{t}$$

$$|\sum_{x=1}^{50} e(tx^2/500)|$$



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Some peaks bigger than 12 or so. Random noise ≤ 12 .

Repulsion: pick points $t, t + \beta$. If both are at peaks, $|\beta| < 1$ or $|\beta| > 4$.

So each peak has width 1, and they are at least 4 apart. Consequently the measure of t lying on peaks is small.

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- $\vec{f}$ is **smooth** if $\vec{f} = 0$ defines a smooth $s - R$ dimensional complex manifold in $\mathbb{C}^s \setminus \{\vec{0}\}$ (away from the origin).

Theorem (Birch 1962)

We have $N_{\vec{f}}(B) \sim c_{\vec{f}} B^{s-dR}$ as above if $\vec{f}$ **smooth**, $d_1 = \dots = d_R = d$,
 $s \geq (d-1)2^{d-1}R(R+1) + R$.

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Generalisations to unequal d_i .

- Browning–Dietmann–Heath-Brown (2014), $d_1 = 2, d_2 = 3, s \geq 29$.
- If $d_1, \dots, d_R \leq D$, Browning–Heath-Brown (2017) handle around $R^2 D^2 2^{D-1}$ variables, improving on Schmidt (1985).

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Notation

- $\vec{g}(\vec{x}) \in \mathbb{R}[\vec{x}]^R$ is a system of R forms in s variables of degrees $d_i \geq 2$.
 - The function $M(B) := \#\{\vec{x} \in \mathbb{Z}^s \setminus \{\vec{0}\} : \|\vec{g}(\vec{x})\| \leq 1, \|\vec{x}\| \leq B\}$.
 - $\vec{g}$ is nonsingular if the $R \times s$ Jacobian matrix $(\partial g_i(\vec{x})/\partial x_j)_{ij}$ has rank R at every nontrivial complex solution $\vec{x}$ to $\vec{g}(\vec{x}) = \vec{0}$.
 - $\vec{g}(\vec{x})$ is **irrational** if no $\vec{\alpha} \in \mathbb{R}^R \setminus \{\vec{0}\}$ satisfies $\vec{\alpha} \cdot \vec{g}(\vec{x}) \in \mathbb{Z}[\vec{x}]$.
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- Case of Bentkus and Götze (1999): circle method proof of Eskin, Margulis and Mozes' formula $M(B) \sim \nu_g B^{s-2}$ in case when **s** $\geq$ **9**.

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Theorem (Müller, 2008, slightly rephrased)

If $d_1 = \dots = d_R = 2$, $\vec{g}$ is nonsingular and irrational, and $s \geq 9R$, then we have $M(B) \sim \nu_{\vec{g}} B^{s-2R}$ as $B \rightarrow \infty$ for some $\nu \geq 0$.

- Freeman (2000, 2001): Standardised this form of the circle method.
- Buterus-Götze-Hille-Margulis (2022): EMM's $R = 1$, $s \geq 5$, explicit, by circle method! Exp sums $\ll$ nice functions on 1-param groups

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- Coefficients in $\mathbb{Q}$, in a degree m number field K or in $\mathbb{R}$. **No averaging over $\vec{a}$ with $\vec{a} \approx \vec{a}/q$.**
- Show $\sum_{\vec{x}: \vec{f}=\vec{0}} w(\vec{x}/B) \sim \mathcal{O} B^{m(s-|\vec{d}|)}$ for $\mathbb{Q}$ or K , or $\sum_{\vec{x}: |\vec{f}| \leq 1/2} w(\vec{x}/B) \sim \mathcal{I} P^{n-|\vec{d}|}$ for $\mathbb{R}$.
- Assume only a nonsingularity-type condition, an irrationality condition if working over $\mathbb{R}$, and a lower bound on s . In the number field case, the bound on s must be independent of K .

$$\begin{array}{c} \text{Birch 1962, over } \mathbb{Q} \\ R(R+1)(d-1)2^{d-2} \\ +R \end{array}$$

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 $d=4$ over $\mathbb{Q}$
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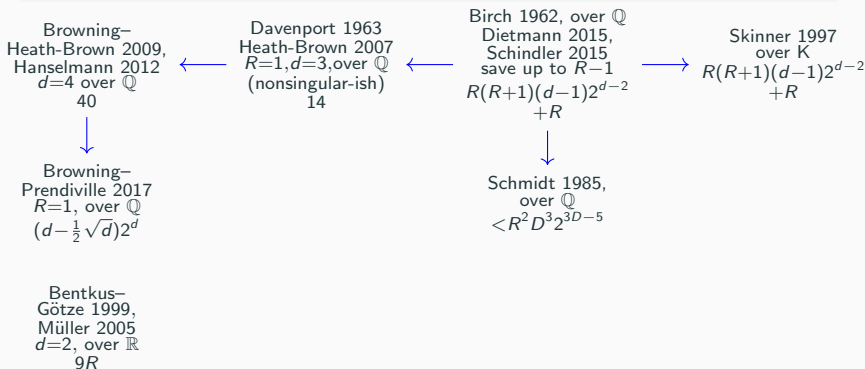


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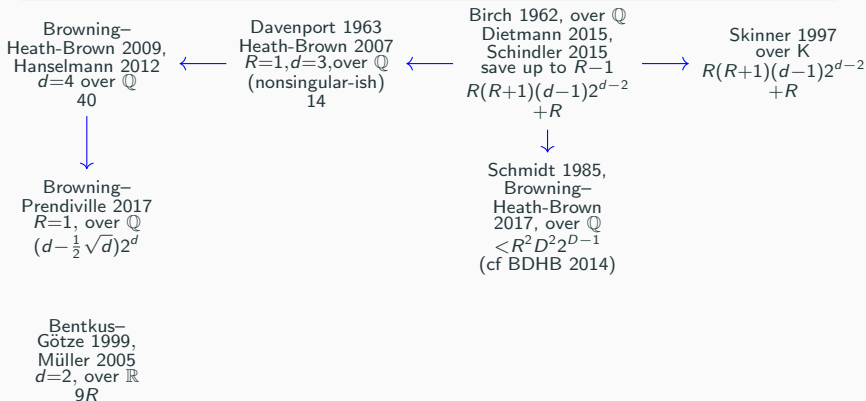
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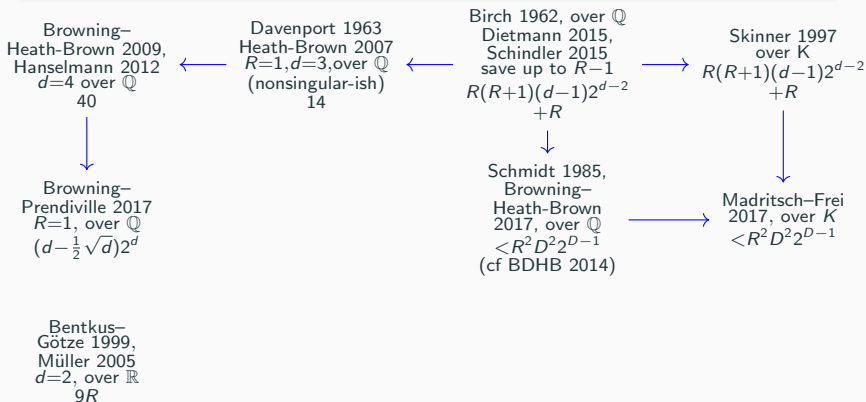
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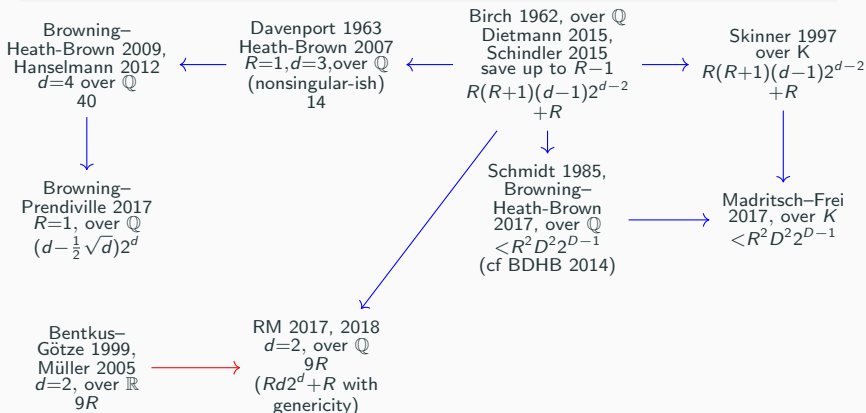
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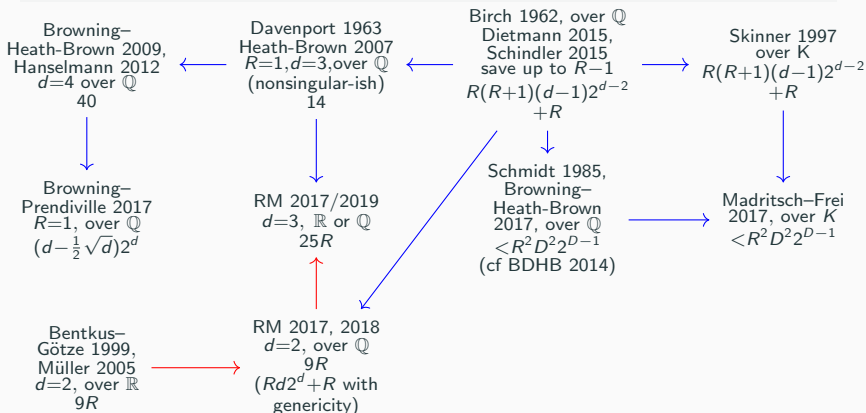
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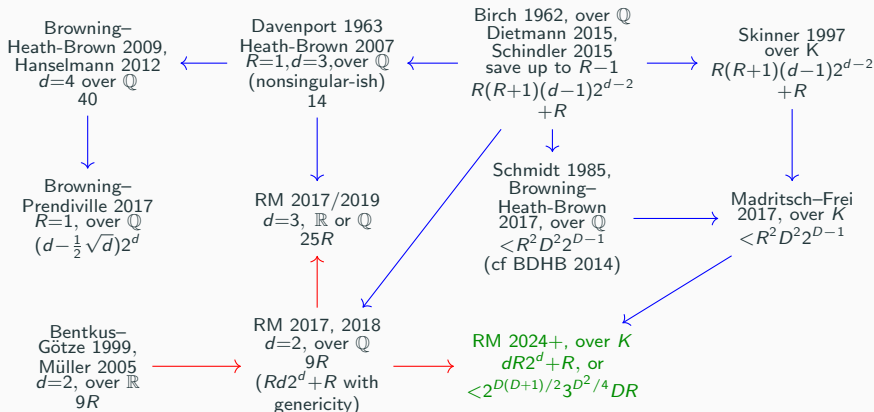
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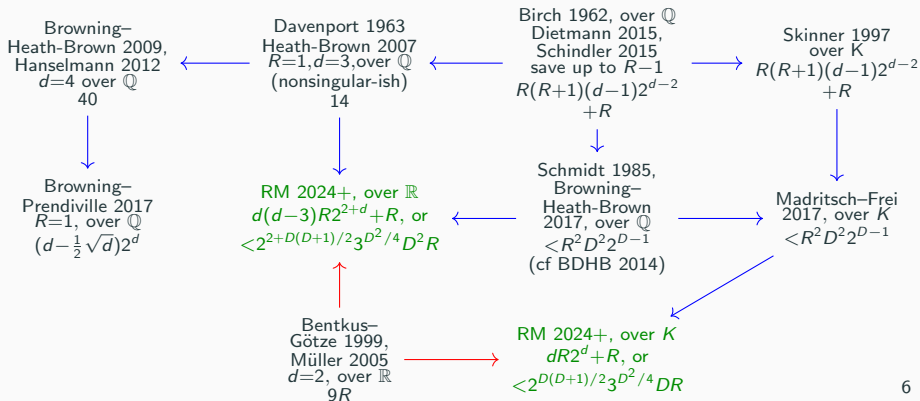
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- Assume only a nonsingularity-type condition, an irrationality condition if working over $\mathbb{R}$, and a lower bound on s . In the number field case, the bound on s must be independent of K .



The rules

- $\vec{f}$ is R forms in s integer variables $\vec{x}$; degree $d_1 = \dots = d_R = d$ or $2 \leq d_i \leq D$; $|\vec{d}| = \sum d_i$.
- Coefficients in $\mathbb{Q}$, in a degree m number field K or in $\mathbb{R}$. **No averaging over $\vec{a}$ with $\vec{a} \approx \vec{a}/q$.**
- Show $\sum_{\vec{x}: \vec{f}=\vec{0}} w(\vec{x}/B) \sim \mathcal{O} B^{m(s-|\vec{d}|)})$ for $\mathbb{Q}$ or K , or $\sum_{\vec{x}: |\vec{f}| \leq 1/2} w(\vec{x}/B) \sim \mathcal{O} P^{n-|\vec{d}|}$ for $\mathbb{R}$.
- Assume only a nonsingularity-type condition, an irrationality condition if working over $\mathbb{R}$, and a lower bound on s . In the number field case, the bound on s must be independent of K .



Theorem (RM 2024+)

$$N_{\vec{f}}(B) \sim c_{\vec{f}} B^{m(s-|\vec{d}|)} \text{ if } \vec{f} = 0 \text{ smooth, } R_d = \#\{d_i \geq d\}, r_d = \#\{d_i = d\},$$

$$\frac{(s+1-R_D) \min\{1, 2D/(r_D+1)(D-1)\}}{2^D} + \sum_d \frac{(s+1-R_d) \min\{1, 2d/(r_d+1)(d-1)\}}{(2^{D-1}+1)\dots(2^d+1)2^d 3^{(d+1)(D-d-1)+1}} > |\vec{d}|.$$

Typically this is about $2^{D(D+1)/2} 3^{d^2/4} DR$ variables; when $d_1 = \dots = d_R = d$ it is $(1 + d2^d)R$ variables.

Ideas: p -adic repulsion ($g \not\equiv h \pmod{\mathcal{O}_K[\vec{x}]} \Rightarrow |\sum e(f(\vec{x})) \sum e(g(\vec{x}))| < |g - h|_p^C$), accessing low degrees in $\sum e(f(\vec{x}))$ (“reiterative differencing”).

Theorem (RM 2024+)

$N_{\vec{f}}(B) \sim c_{\vec{f}} B^{m(s-|\vec{d}|)}$ if $\vec{f} = 0$ smooth, $R_d = \#\{d_i \geq d\}$, $r_d = \#\{d_i = d\}$,

$$\frac{(s+1-R_D) \min\{1, 2D/(r_D+1)(D-1)\}}{2^D} + \sum_d \frac{(s+1-R_d) \min\{1, 2d/(r_d+1)(d-1)\}}{(2^{D-1}+1) \dots (2^d+1) 2^d 3^{(d+1)(D-d-1)+1}} > |\vec{d}|.$$

Typically this is about $2^{D(D+1)/2} 3^{d^2/4} DR$ variables; when $d_1 = \dots = d_R = d$ it is $(1 + d2^d)R$ variables.

Ideas: p-adic repulsion ($g \not\equiv h \pmod{\mathcal{O}_K[\vec{x}]} \Rightarrow |\sum e(f(\vec{x})) \sum e(g(\vec{x}))| < |g - h|_p^C$), accessing low degrees in $\sum e(f(\vec{x}))$ (“reiterative differencing”).

Theorem (RM 2024+)

$s'_d = s - R_d$ ($d \leq 3$) or $s'_d = \lfloor \frac{s+1-R_d}{2(d-3)} + \frac{1}{2} \rfloor$, $s'_d = \frac{s_d^{(1)}}{2} (1 - \frac{(d-3)(s_d^{(1)}-1)}{s+1-R_d})$ ($d \geq 4$). Suppose that $\vec{g}$ is nonsingular and irrational, and that

$$\frac{s'_D}{2^D} + \sum_d \frac{s'_d}{(2^{D-1}+1) \dots (2^d+1) 2^d 3^{(d+1)(D-d-1)+1}} > |\vec{d}|$$

Then $M(B) \sim \nu_{\vec{g}} B^{n-\sum d_i}$ where $\nu_{\vec{g}} = \lim B^{\sum d_i} \mathbb{P}[\vec{g}(\vec{X}) \in [0, 1)^R]$.

$s'_d \approx s/3(d-3)$; about $8(D-3)$ times as many variables. Ideas: reiterative differencing, work of Davenport, very pedantic book-keeping.

Setup for repulsion

- $e(t) = 2^{2\pi it}$, $\Delta_{\vec{h}} f(\vec{x}) = f(\vec{x} + \vec{h}) - f(\vec{x})$, $\Delta_{\vec{h}_1, \dots, \vec{h}_r} = \Delta_{\vec{h}_1} \cdots \Delta_{\vec{h}_r}$
- $f^{[k]}$ is the degree k part of f , with $|f^{[k]}| = |\text{biggest coefficient}|$.
- $\vec{m}_{\vec{h}_1, \dots, \vec{h}_{k-1}}^{f, k}$ is the vector of coefficients of $\Delta_{\vec{h}_1, \dots, \vec{h}_{k-1}, (\cdot)} f^{[k]}$
- The *normalised* sum $S_w(B; f) = \sum_{\vec{x} \in \mathbb{Z}^s} B^{-s} w(\vec{x}/B) e(f(\vec{x}))$

For $f, g \in \mathbb{R}[\vec{x}]$ with degree $\leq k$, and $pB^{-k} \ll |f^{[k]}| \ll p^{1-k}$ we have

$$|S_w(B; f) S_w(B; g + f)|^{2^{k-1}} \ll p^{-(k-1)s} \cdot \#\{(\vec{h}_1, \dots, \vec{h}_{k-1}) \in \mathbb{Z}^{(k-1)s} : |\vec{h}_i| \leq p, |\vec{m}_{\vec{h}_1, \dots, \vec{h}_{k-1}}^{f, k}| \leq p^{k-2} |f^{[k]}|\}.$$

If $k = 3$, $f^{[3]}$ smooth get $\ll p^{-s}$ by ideas of Davenport; else, pigeonhole.

Repulsion: suppose that, whenever $pB^{-k} \leq |\vec{\beta} \cdot \vec{f}^{[k]}| \leq p^{1-k}$, we have

$$S_w(B; \vec{\alpha} \cdot \vec{f}) S_w(B; (\vec{\alpha} + \vec{\beta}) \cdot \vec{f}) \ll p^{-2E(k)},$$

then $\text{meas}\{\vec{\alpha} \in [0, 1)^R : |S_w(B; \vec{\alpha} \cdot \vec{f})| \asymp B^{-A}\} \ll B^{\sum^R d_i(A/E(d_i)-1)}$ ($A > 0$)

○: If $\sum^R d_i/E(d_i) < 1$ then $\begin{cases} N_{\vec{f}}(B) \sim c_{\vec{f}} B^{s - \sum^R d_i} & \text{if } \vec{f} \in \mathbb{Z}[\vec{x}]^R, \text{ or} \\ M(B) \sim \nu_{\vec{f}} B^{s - \sum^R d_i} & \text{if } \vec{f} \text{ is irrational.} \end{cases}$

Setup for p -adic repulsion

- $e(t) = 2^{2\pi it}$, $\Delta_{\vec{h}} f(\vec{x}) = f(\vec{x} + \vec{h}) - f(\vec{x})$, $\Delta_{\vec{h}_1, \dots, \vec{h}_r} = \Delta_{\vec{h}_1} \cdots \Delta_{\vec{h}_r}$
- $f^{[k]}$ is the degree k part of f , with $|f^{[k]}| = |\text{biggest coefficient}|$.
- $\vec{m}_{\vec{h}_1, \dots, \vec{h}_{k-1}}^{f, k}$ is the vector of coefficients of $\Delta_{\vec{h}_1, \dots, \vec{h}_{k-1}, (\cdot)} f^{[k]}$
- The *normalised* sum $S_w(B; f) = \sum_{\vec{x} \in \mathbb{Z}^s} B^{-s} w(\vec{x}/B) e(f(\vec{x}))$

For $f \in \mathbb{Z}[\vec{x}]$, $g \in \mathbb{R}[\vec{x}]$ with degree $\leq k$, and p prime with $p^M \ll B^k$, such that $p^{-M} < |f^{[k]}|_p \leq p^{1-k}$ we have

$$|S_w(B; f) S_w(B; g + \frac{1}{p^M} f)|^{2^{k-1}} \ll p^{-(k-1)s} \cdot \#\{(\vec{h}_1, \dots, \vec{h}_{k-1}) \in \mathbb{Z}^{(k-1)s} : |\vec{h}_i| \leq p, |\vec{m}_{\vec{h}_1, \dots, \vec{h}_{k-1}}^{f, k}|_p \leq p^{-1} |f^{[k]}|_p\}.$$

For $f^{[k]}$ smooth this is $\ll p^{-s}$ ($\mathbb{F}_p$ -points on varieties).

Repulsion: suppose that, whenever $p^{-M} < |\vec{b} \cdot \vec{f}^{[k]}|_p \leq p^{1-k}$, we have

$$S_w(B; \vec{\alpha} \cdot \vec{f}) S_w\left(B; \left(\vec{\alpha} + \frac{\vec{b}}{p^M}\right) \cdot \vec{f}\right) \ll p^{-2E(k)},$$

then $\text{meas}\{\vec{\alpha} \in [0, 1]^R : |S_w(B; \vec{\alpha} \cdot \vec{f})| \asymp B^{-A}\} \ll B^{-\sum^R d_i(A/E(d_i)-1)}$ ($A > 0$)

○: If $\sum^R d_i/E(d_i) < 1$ then $N_{\vec{f}}(B) \sim cB^{s-\sum^R d_i}$. Smooth: $E(d) = \frac{n-R+1}{2^d}$. 8

Measures by repulsion

- $e(t) = 2^{2\pi it}$, $\Delta_{\vec{h}} f(\vec{x}) = f(\vec{x} + \vec{h}) - f(\vec{x})$, $\Delta_{\vec{h}_1, \dots, \vec{h}_r} = \Delta_{\vec{h}_1} \cdots \Delta_{\vec{h}_r}$
- $f^{[k]}$ is the degree k part of f , with $|f^{[k]}| = |\text{biggest coefficient}|$.
- $\vec{m}_{\vec{h}_1, \dots, \vec{h}_{k-1}}^{f, k}$ is the vector of coefficients of $\Delta_{\vec{h}_1, \dots, \vec{h}_{k-1}, (\cdot)} f^{[k]}$
- The *normalised* sum $S_w(B; f) = \sum_{\vec{x} \in \mathbb{Z}^s} B^{-s} w(\vec{x}/B) e(f(\vec{x}))$
- $\vec{f}(\vec{x}) \in \mathbb{Z}[x_1, \dots, x_s]^R$ is a system of R forms in $s > \sum d_i$ variables.

Repulsion: suppose that, whenever $pB^{-k} \leq |\vec{\beta} \cdot \vec{f}^{[k]}| \leq p^{1-k}$, we have

$$S_w(B; \vec{\alpha} \cdot \vec{f}) S_w(B; (\vec{\alpha} + \vec{\beta}) \cdot \vec{f}) \ll p^{-2E(k)},$$

then $\text{meas}\{\vec{\alpha} \in [0, 1)^R : |S_w(B; \vec{\alpha} \cdot \vec{f})| \asymp B^{-A}\} \ll B^{\sum^R d_i(A/E(d_i)-1)}$ ($A > 0$)

proof idea: Let $p \gg B^{A/E(k)}$. Let $\vec{\alpha}, \vec{\alpha} + \frac{\vec{b}}{p^M}$ belong to the set. Then

$$|\vec{\beta} \cdot \vec{f}^{[k]}| < pB^{-k}, \text{ or } |\vec{\beta} \cdot \vec{f}^{[k]}| > p^{1-k}.$$

It follows that the $\vec{\alpha} \cdot \vec{f}^{[k]}$ are contained in a few infrequent regions (peaks) of diameter $\leq pB^{-k}$, separated by gaps of size $\geq p^{1-k}$, hence with total measure $\leq (p/B)^{kR_k}$ where there are R_k forms of degree k .

Measures by p -adic repulsion

- $e(t) = 2^{2\pi it}$, $\Delta_{\vec{h}} f(\vec{x}) = f(\vec{x} + \vec{h}) - f(\vec{x})$, $\Delta_{\vec{h}_1, \dots, \vec{h}_r} = \Delta_{\vec{h}_1} \cdots \Delta_{\vec{h}_r}$
- $f^{[k]}$ is the degree k part of f , with $|f^{[k]}| = |\text{biggest coefficient}|$.
- $\vec{m}_{\vec{h}_1, \dots, \vec{h}_{k-1}}^{f, k}$ is the vector of coefficients of $\Delta_{\vec{h}_1, \dots, \vec{h}_{k-1}, (\cdot)} f^{[k]}$
- The *normalised* sum $S_w(B; f) = \sum_{\vec{x} \in \mathbb{Z}^s} B^{-s} w(\vec{x}/B) e(f(\vec{x}))$
- $\vec{f}(\vec{x}) \in \mathbb{Z}[x_1, \dots, x_s]^R$ is a system of R forms in $s > \sum d_i$ variables.

Repulsion: suppose that, whenever $p^{-M} < |\vec{b} \cdot \vec{f}^{[k]}|_p \leq p^{1-k}$, we have

$$S_w(B; \vec{\alpha} \cdot \vec{f}) S_w\left(B; \left(\vec{\alpha} + \frac{\vec{b}}{p^M}\right) \cdot \vec{f}\right) \ll p^{-2E(k)},$$

then $\text{meas}\{\vec{\alpha} \in [0, 1)^R : |S_w(B; \vec{\alpha} \cdot \vec{f})| \asymp B^{-A}\} \ll B^{\sum^R d_i(A/E(d_i)-1)}$ ($A > 0$)

proof idea: Let $p \gg B^{A/E(k)}$ prime, $p^M \ll B^k$. Let $\vec{\alpha}, \vec{\alpha} + \frac{\vec{b}}{p^M}$ belong to the set. Either $\vec{\beta} = 0$ or $\vec{\beta} \in p^{k-2-M}\mathbb{Z}$, because

$$|\vec{b} \cdot \vec{f}^{[k]}|_p \leq p^{-M}, \text{ or } |\vec{b} \cdot \vec{f}^{[k]}|_p > p^{1-k}.$$

Hence each lattice $\vec{\alpha}_0 + \frac{1}{p^{k-1-M}}\mathbb{Z}^R$ contains at most one element of the set, hence it has total measure $\leq p^{(k-1-M)R_k}$ where there are R_k forms of degree k , this is $\leq (p/B)^{kR_k}$ if we choose M so $B^{-k} \gg p^{-M-1}$.

Notation

- $e(t) = 2^{2\pi it}$, $\Delta_{\vec{h}} f(\vec{x}) = f(\vec{x} + \vec{h}) - f(\vec{x})$, $\Delta_{\vec{h}_1, \dots, \vec{h}_r} = \Delta_{\vec{h}_1} \cdots \Delta_{\vec{h}_r}$
- NB $\Delta_{\vec{h}_1, \dots, \vec{h}_k} f^{[k]}$ is a multilinear form in the $\vec{h}_i$, independent of $\vec{x}$
- $\vec{m}_{\vec{h}_1, \dots, \vec{h}_{k-1}}^{f, k}$ is the vector of coefficients of $\Delta_{\vec{h}_1, \dots, \vec{h}_{k-1}, (\cdot)} f^{[k]}$

Proposition

Given $w \in C_c^\infty(\mathbb{R}^s)$ there is $\tilde{w} \in C_c^\infty(\mathbb{R}^{ks})$ as follows. For any $f \in \mathbb{R}[\vec{x}]$ of degree $\leq d$, and any $1 < k < d$,

$$\begin{aligned}
 & \left| \sum_{\vec{x} \in \mathbb{Z}^s} \frac{1}{B^s} w\left(\frac{\vec{x}}{B}\right) e(f(\vec{x})) \right| (2^{d-1} + 1) \cdots (2^k + 1) 2^{k-1} 3^{(k+1)(d-k-1)+1} \\
 & \leq B^{-ks} \sum_{\vec{h}_1, \dots, \vec{h}_{k-1} \in \mathbb{Z}^s} \left| \sum_{\vec{h}_k \in \mathbb{Z}^s} \tilde{w}\left(\frac{\vec{h}_1}{B}, \dots, \frac{\vec{h}_k}{B}\right) e\left(\Delta_{\vec{h}_1, \dots, \vec{h}_k} f^{[k]}\right) \right| \\
 & \ll \frac{1}{B^{(k-1)s}} \#\{(\vec{h}_1, \dots, \vec{h}_{k-1}, \vec{u}) \in \mathbb{Z}^{ks} : |\vec{h}_i| \leq B, |\vec{m}_{\vec{h}_1, \dots, \vec{h}_{k-1}}^{f, k} - \vec{u}| \leq \frac{1}{B}\}
 \end{aligned}$$

Lemma

Given $w \in C_c^\infty(\mathbb{R}^s)$ there is $\tilde{w} \in C_c^\infty(\mathbb{R}^{(d+1)s})$ as follows. For any $f \in \mathbb{R}[\vec{x}]$ of degree $\leq d$,

$$\left| \sum_{\vec{x} \in \mathbb{Z}^s} \frac{1}{B^s} w\left(\frac{\vec{x}}{B}\right) e(f(\vec{x})) \right|^{2^{d-1} + 1} \\ \leq \left| \sum_{\vec{x}_1, \dots, \vec{x}_d} \sum_{\vec{x}} \tilde{w}\left(\frac{\vec{x}_d}{B}, \dots, \frac{\vec{x}_1}{B}, \frac{\vec{x}}{B}\right) e\left(f^{[<d]} + F(\vec{x}; \vec{x}_1, \dots, \vec{x}_d)\right) \right|$$

with $F = f^{[d]} - \Delta_{\vec{x} + \vec{x}_1, \dots, \vec{x} + \vec{x}_d} f^{[d]}$ of degree $< d$ in $\vec{x}$ and ≤ 1 in each $\vec{x}_i$.

Lemma

For $L \in \text{GL}_s(\mathbb{R})$, $A := \{\|\vec{x}\| \leq B : \|L\vec{x} - \vec{u}\| \leq 1/B \text{ some } \vec{u} \in \mathbb{Z}^s\}$.

Given $w \in C_c^\infty(\mathbb{R}^{2s})$ there are $\tilde{w}_{\vec{c}}^{L,B} \in C_c^\infty(\mathbb{R}^s)$, whose values, support and derivatives are bounded in terms of w only, such that for all real g ,

$$\left| \sum_{\vec{x}, \vec{y}} w\left(\frac{\vec{x}}{B}, \frac{\vec{y}}{B}\right) e(\vec{y} \cdot L\vec{x} + g(\vec{x})) \right|^3 \leq \sum_{\vec{c} \in A-A} \sum_{\vec{x}} \tilde{w}_{\vec{c}}^{L,B}\left(\frac{\vec{x}}{B}\right) e(\Delta_{\vec{c}} g(\vec{x})).$$